you. DNA or Deoxyribonucleic Acid is incredibly complex and we have yet to scratch the surface of all the secrets it holds within it.

CURRENT DIFFICULTIES IN TESTING

Testing for those 50 genetic diseases is especially challenging since the symptoms are extremely complex. Also, the nature of repetitive sequences combined with the limitations of existing genetic testing methods makes it difficult as well. Currently, doctors need to test based on the symptoms and a thorough evaluation of medical conditions of the family history and even then, many times the results could be inconclusive which can cause additional stress to the patient and their family members. This testing process can often go on for years without finding the genes responsible for the cause of the disease. In medical terms, this lack of clear diagnosis is referred to as 'diagnostic odyssey'. When it comes to repeat expansion disorders, it is imperative that the correct disease be diagnosed as early as possible. Although these diseases are incurable, early diagnosing can help with treating the disease earlier and prevent complications that may arise as time goes by. And that is why this new single test system can be a life-changing boon for patients with genetic diseases. This test will also help prevent years of unneeded muscle and nerve biopsies for the plethora of diseases the patient does not have and will help prevent the use of risky treatment measures that may suppress the patient's immune system.

BENEFITS OF NEW TEST

In one fell swoop, testing for these 50-plus genetic disorders which required complicated biopsies has been reduced to a simple blood-test like procedure. Not only has it become quick and easy, but it is also extremely accurate. Ira Deveson, Head of Genomics Technologies at the Garvan Institute in Sydney said, "We correctly diagnosed all patients with conditions that were already known, including Huntington's disease, fragile X syndrome, hereditary cerebellar ataxias, myotonic dystrophies, myoclonic epilepsies, motor neuron disease and more,".

This new test is all set to revolutionise how these genetic diseases are diagnosed. All the disorders can now be tested for with just a single test which is light years ahead of the tests that could take years to diagnose and multiple tests needed to be conducted for each suspected disease. This test works using a technology called Nanopore sequencing and works by using the DNA sample of the patient, usually extracted from the blood. The long repeated DNA sequences that cause the aforementioned diseases are read by the Nanopore device which is designed to hone in on the roughly 40 genes that malfunction.

This one test can be used to search for every known disease-causing repeat expansion sequence and not only does it make it easy to diagnose known diseases, but it can also be used to discover novel sequences that would likely be involved in causing diseases but have yet to be described.

The benefits of this new test don't just stop at better and more efficient diagnoses. The Nanopore gene sequencing device is more compact, being as small as a stapler and costs just around $1000. Compare that to the hundreds of thousands of dollars that were previously needed for DNA sequencing technologies that ended up being the size of a fridge, and the benefits are obvious.

FUTURE OF NANOPORE TECHNOLOGY

With these revolutionary findings the research team from Garvan Institute of Medical Research along with collaborators from Australia, UK and Isreal have their eyes set on bringing this new technology for diagnostic practice in the next two to five years. The next step on their way to making Nanopore testing mainstream is to get this process and technology clinically accredited which is in essence a self-assessment and external peer assessment of the technology and compare the performance of the new method when compared to the traditional and established testing methods. This process is used to iron out bugs (if there are any) in the testing method and it can even be used to improve the process.

Considering that adult-onset genetic disorders have not been given as much priority when it comes to research into the subject, this new testing method will revolutionise research into genetic diseases. It can be used to diagnose people with these rare genetic diseases even before any symptoms appear. Which, of course, means that the treatment can be started way in advance reducing the brunt of the disorders in the future. d

"Current genetic testing for expansion disorders can be hit and miss. When patients present with symptoms, it can be difficult to tell which of these 50-plus genetic expansions they might have, so their doctor must decide which genes to test for based on the person's symptoms and family history. If that test comes back negative, the patient is left without answers. This testing can go on for years without finding the genes implicated in their disease. We call this the 'diagnostic odyssey', and it can be quite stressful for patients and their families,"

— DR KISHORE KUMAR, Co-author of the study. Clinical neurologist, Concord Hospital.